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StyleNeRF-Med: A hybrid neural framework for robust 3D image reconstruction from sparse 2D views

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HIGHLIGHTS

- Proposes a hybrid deep learning framework (StyleNeRF-Med) combining StyleGAN2 and Neural Radiance Fields for 3D image reconstruction.
- Demonstrates robust 3D reconstruction from sparse 2D input views without requiring dense sampling or precise camera calibration.
- Integrates generative and volumetric modeling to tolerate data incompleteness-aligning with soft computing principles.
- Outperforms existing NeRF-based methods in reconstruction quality using standard image quality metrics (FID and KID).
- Offers a data-efficient solution applicable to medical imaging, robotics, and other real-world domains with limited visual data.

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ABSTRACT

Reconstructing high-quality 3D images from fewer input images is a fundamental challenge in computer vision and medical imaging, which has significant implications for data-efficient artificial intelligence. Traditional methods such as computed tomography and magnetic resonance imaging typically require multiple images for accurate image reconstruction, making them costly and data-intensive. Recent advances in the field of neural radiance fields (NeRF) have shown strong performance in the development of novel views. However, such methods are highly dependent on precise camera settings and scene-specific optimization, which limits their generalizability and usefulness. To overcome these limitations, we leverage advanced AI techniques, such as Generative Adversarial Networks (GANs) for generating realistic images. We propose StyleNeRF-Med, a hybrid framework that innovatively integrates StyleGAN2 and NeRF to generate high-quality 3D reconstructions from fewer input images. StyleGAN2 encodes rich image features to synthesize detailed views, while NeRF processes 3D spatial coordinates and camera positions to compute accurate pixel colors and intensities. A synthetically generated dataset of digitally reconstructed radiographs (DRR) is used to validate the model. StyleNeRF-Med demonstrates superior performance in reconstructing high-quality 3D medical images from limited 2D input. On the chest dataset, it achieves the lowest FID score of 101.24 and KID score of 0.072, outperforming GRAF and PixelNeRF. Similarly, on the knee dataset, StyleNeRF-Med again shows better image quality with the lowest FID score of 132.64 and the KID score of 0.098. These results indicate improved alignment with real data distributions and improved reconstruction quality over existing methods.

1. Introduction

Medical imaging is a key tool for the diagnosis, monitoring, and treatment of diseases in modern healthcare. Despite their important role, conventional medical imaging techniques such as computed tomography (CT), magnetic resonance imaging (MRI) and X-rays often face challenges due to image resolution, noise, and the ability to effectively visualize three-dimensional structures [1]. These limitations may lead to ambiguities and uncertainties in diagnosis. Recent research [2,3] highlights the impact of such limitations on clinical outcomes and emphasizes the importance of improved imaging techniques. The development of improved imaging technologies that can overcome such

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limitations is therefore crucial for both clinical practice and medical research [4,5].

Image reconstruction techniques are widely used in healthcare to provide advanced diagnostic capabilities [6]. In medical diagnostics, it is important to accurately reconstruct the internal anatomy from a few medical images (X-rays or scans). Traditional imaging techniques used in medical diagnostics, such as CT and magnetic resonance imaging, typically require multiple images to produce high-quality images. However, acquiring multiple input images can lead to prolonged exposure to ionizing radiation, which is hazardous to the patient's health. Therefore, advanced imaging techniques are needed that can produce novel high-quality images from a very limited number of images to improve healthcare diagnostics.

Existing approaches to image reconstruction in healthcare are mostly based on analytical and iterative methods for given input data. Such approaches often struggle with mismatches between mathematical models and the physical characteristics of imaging devices [7]. Several state-of-the-art approaches that utilize deep learning [8,9] have recently become popular in sparse view reconstruction [10,11] and 3D CT reconstruction of 2D images [12]. Such DL-based approaches optimize modern architectures to overcome the challenge of mismatches between the mathematical model and the imaging system and produce improved reconstructions. However, such advanced DL approaches require a large amount of training data, which is difficult to obtain in the medical domain.

Neural Radiance Fields (NeRF) is a popular approach in image reconstruction that generates volumetric 3D representations from input images [13]. The parameters of a neural network are used to learn how to generate new views by volumetric rendering from point patterns along rays traversing the image to capture its radiance field and density. Despite its wide popularity for the synthesizing novel views using volumetric rendering, NeRF struggles to reconstruct medical images due to the dynamic characteristics of clinical environments, the need for longer imaging sessions, and the complexity of overlapping anatomical structures. These limitations restrict the applicability of NeRF in the medical field.

Generative Adversarial Networks (GANs) have emerged as a powerful paradigm for image reconstruction [14,15]. Models trained with GANs can learn the underlying distribution of data for image reconstruction. Using learned priors, GANs can reconstruct images by synthesizing new views from fewer input data. Despite recent advances in NeRF-GAN hybrid models such as GRAF, Pi-GAN, EG3D, and HyperNeRFGAN, their effectiveness in medical imaging remains limited. These models are primarily designed for natural images and assume well-defined textures, dense multi-view data, or reliable camera pose estimation. In contrast, medical imaging involves sparse views, low contrast, noise, and overlapping anatomical structures, which degrade reconstruction quality. Furthermore, existing NeRF-based medical approaches lack strong generative priors for fine detail recovery, while GAN-based models do not explicitly model volumetric density and attenuation properties. This highlights a critical gap in developing a unified framework tailored to medical imaging constraints. To address the limitations of current medical image reconstruction, we propose to integrate StyleGAN2 with NeRF to reconstruct high-quality medical images. We chose StyleGAN2 due to its ability to handle complex textures and details that are essential for the accurate rendering of medical images. NeRF, on the other hand, is used for its ability to produce precise 3D volumetric outputs from sparse image data. Consequently, this work proposes StyleNeRF-Med, a novel integration of StyleGAN2 and NeRF to generate novel 3D image reconstructions from sparse 2D views.

This paper proposes StyleNeRF-Med, a hybrid methodology that integrates the strengths of StyleGAN2 and NeRF to reconstruct 3D medical images. The proposed model integrates the latent-space representation of structural features in StyleGAN2 with the volumetric rendering capabilities of NeRF. This innovative integration enables it to produce anatomically significant reconstructions from a limited 2D input images. Unlike other existing approaches, StyleNeRF-Med implements a style-controlled latent modulation within a NeRF pipeline, enhancing the model's capacity to generalize across patient variability. This distinguishes the proposed approach from existing NeRF-GAN models, which are not explicitly designed to handle the unique challenges of medical imaging data. We also demonstrate its scalability beyond medical imaging, showing that StyleNeRF-Med can accommodate non-medical applications while maintaining structural and visual consistency.

The proposed StyleNeRF-Med aims to address existing challenges in medical image reconstruction by reducing the reliance on larger input datasets and improving visualization of anatomical overlaps. Unlike existing GAN-NeRF hybrid models that primarily focus on general scene synthesis, the proposed StyleNeRF-Med introduces a domain-adaptive integration specifically designed for medical imaging. It leverages StyleGAN2 for structured latent space learning and combines it with volumetric NeRF rendering to handle sparse-view reconstruction and overlapping anatomical structures. Additionally, the framework is tailored to operate effectively on projection-based medical data (DRRs). The specific contributions of this paper are as follows:

- We propose to integrate StyleGAN2 with the NeRF to reconstruct high-quality medical images and design StyleNeRF-Med, which generates novel views for medical images.
- We implement StyleNeRF-Med and demonstrate its ability to generate novel views for medical image reconstruction with fewer input data.
- We comparatively evaluate StyleNeRF-Med against other popular models like PixelNeRF and GRAF using synthetically generated digital reconstructed radiographs to show the capability of the proposed model in generating high quality novel views.

The remainder of the paper is organized as follows: Section 2 provides a brief review of current state-of-the-art approaches for image synthesis. Section 3 briefly describes the architecture and workings of the proposed model followed by the implementation methodology in Section 4. Section 5 provides the results obtained and their evaluation. Lastly, Section 6 concludes the paper.

2. Literature review

In this section, we provide a comprehensive review of popular existing models.

2.1. NeRF-based approaches

Trevithick et al. [16] propose the general radiation field (GRAF) for 3D representation. The GRAF model helps to show 3D scenes in 3D using multi-view geometry. It accomplishes this by encoding scenes as radiance fields, which facilitate high-quality rendering of 3D images from 2D images. GRAF can adapt to complex scenes and achieve multi-view consistency, though it requires multiple views for accuracy. Yuan et al. [17] propose an interactive NeRF that integrates shape priors with a NeRF architecture to enable user-controlled modification of 3D models. The proposed interactive NeRF enables intuitive shape deformation through external 3D modeling tools by deriving a mesh representation. This method is effective for interactive editing, but its applicability is limited to static scenes and is computationally expensive. PixelNerf proposed by Yu et al. [18] improves NeRF by employing convolutional conditioning to generate real-time 3D views with minimal input data by conditioning the synthesis of 3D scenes on one or a few images. The proposed PixelNerf is particularly effective for the reconstruction of single views and the synthesis of novel views; however, it struggles to handle multi-object scenes. Sitzmann et al. [19] propose Scene Representation Networks (SRN) that provide a continuous neural representation of 3D-structure-aware geometry and appearance without explicit 3D supervision. SRN generalizes well across scenes and maintains multi-view consistency by relying on posed 2D images. Outside pure NeRF pipelines

transformer-based multi-view stereo methods have also improved dynamic context perception and robustness through data augmentation thus demonstrating the continued importance of multi-view feature aggregation for 3D reconstruction [20].

Recent research has also advanced NeRF-based representations in terms of efficiency and scalability. 3D Gaussian Splatting introduces an explicit Gaussian-based scene representation that enables substantially faster training and real-time rendering while maintaining strong novel-view synthesis quality [21]. In contrast, compression-oriented work on Instant-NGP-based NeRF improves storage efficiency by reducing the size of multi-resolution hash embeddings without substantially sacrificing fidelity [22]. Together, these studies show that recent progress is moving toward faster and more compact radiance-field representations, although these methods are not specifically designed for sparse-view medical image reconstruction [23]. NeRF representations have also been extended beyond view synthesis toward 3D scene understanding where channel attention has been used to improve object detection in neural radiance fields [24].

2.2. GAN-based approaches

Hudson et al. [25] propose GANsformer, a transformer with GANs to capture long-range dependencies in image generation. The proposed GANsformer facilitates efficient and scalable image synthesis by employing a bipartite structure. This method is widely used for the synthesis of images and the generation of intricate scenes. Karras et al. [26] propose the StyleGAN architecture which is renowned for its high resolution image synthesis. StyleGAN implements style-based controls and path length regularization for high quality image synthesis. Pix2pix proposed by Isola [27] et al. utilizes conditional GANs for image-to-image translation, demonstrating its adaptability to a variety of tasks, including object reconstruction and colorization. However, it may generate blurry results when trained with Euclidean loss. Barzilay et al. [28] propose a multi-style GAN named MISS that uses a single generator for multiple styles for image-to-illustration translation. The proposed MISS achieves diversity in style retention; however, it is computationally expensive. We provide a brief summary of the existing literature in Table 1.

Chen et al. [32] proposed an efficient geometrically aware 3D GANs (EG3D) image editing framework. The proposed EG3D bridges the gap between photorealistic 2D GANs and 3D scene generation. EG3D integrates StyleGAN2 with a triplane-based explicit-implicit three-dimensional hybrid representation. This triplane architecture encodes information over 3D orthogonally, and employs a lightweight MLP to aggregate and decode this information. The authors [32] also proposed a dual discriminative technique with pose-conditioned generators that ensure multiview consistency throughout the training process. However, EG3D relies on triplane feature representations and requires consistent camera pose information, which limits its applicability in sparse-view medical imaging. In contrast, the proposed StyleNeRF-Med integrates volumetric radiance field modeling with style-based generation, enabling improved reconstruction under limited views without relying on explicit triplane structures.

The EG3D proposed by Chan *et al.* [32] is capable of producing geometrically precise high-resolution, multiview-consistent 3D images. The proposed model separates feature generation via StyleGAN2 from neural rendering, leveraging the strengths of GANs to generate 3D-aware functionalities. Their approach is especially pertinent to the current study, as it demonstrates the integration of StyleGAN2 with volume rendering while maintaining consistency and quality, aligning with the objectives of our proposed StyleNeRF-Med model.

Chan et al. [30] proposed an implicit 3D-aware generative model named pi-GAN. The proposed pi-GAN considers coordinate-based MLPs in conjunction with sinusoidal activations to depict continuous radiance fields. Pi-GAN employs a GAN-based training framework in which the generator acquires implicit scene representations, while the discriminator ensures the authenticity of the rendered 2D images. The proposed model takes a camera pose as input and translates a latent code into RGB values and densities, ultimately producing the image using volume rendering. The strength of pi-GAN lies in its periodic activation functions, which enable the MLP to efficiently capture high-frequency scene

Table 1
Summary of literature on image synthesis approaches.

Model	Architecture	Strengths	Weaknesses	Application Areas
GRAF [16]	NeRF-Based	Multi-view consistency, realistic rendering	Requires multiple images	Robotics, augmented reality
Interactive NeRF [17]	NeRF + Shape priors	User-friendly shape editing	Limited to static scenes	Virtual environments, 3D design
pixelNeRF [18]	NeRF + Conv conditioning	Few-view 3D synthesis	Limited for multi-object scenes	Single-view reconstruction, novel view synthesis
SRN [19]	Continuous 3D network	Multi-view consistency	May struggle with detailed scenes	3D reconstruction, VR, AR
3D Gaussian Splatting [21]	Explicit Gaussian representation	Fast rendering, real-time synthesis	Less robust under difficult inputs	Novel view synthesis, 3D reconstruction
Instant-NGP Compression [22]	Hash-based NeRF compression	Compact storage, preserved fidelity	Not designed for medical imaging	NeRF compression, efficient rendering
GANsformer [25]	Transformer + GAN	Efficient visual synthesis	High computational cost	Complex scene generation, image synthesis
StyleGAN [26]	Style-based GAN	High-fidelity image generation	Artifacts, high computational demand	Photorealistic synthesis, content creation
pix2pix [27]	Conditional GAN	Task adaptability	Produces blurred images occasionally	Image editing, style transfer, object detection
MISS GAN [28]	Multi-style GAN	Diverse style retention	Computationally intensive	Illustration generation, artistic style transfer
HoloGAN [29]	GAN with learned 3D features	Pose control without labels; no need for 3D/pose supervision	Coarse 3D structure; no true volumetric rendering; limited fine detail	Object synthesis and pose manipulation (faces, cars)
Pi-GAN [30]	Implicit radiance field with volume rendering in GAN	Multi-view consistency; sharp details; explicit camera control	Training is computationally heavy; extracted geometry can be noisy	Faces, cats, cars; novel view synthesis from single images
MedNeRF [31]	NeRF-based conditional radiance field	Mdical attenuation modeling; works with sparse views	DRR dependence	Medical CT/DRR projection
EG3D [32]	Tri-plane hybrid explicit–implicit representation	High image quality; strong 3D consistency; fast and memory-efficient	Requires pose estimates/conditioning; not designed for medical imaging	Human/animal faces; general 3D-aware content generation
HyperNeRFGAN [33]	Hypernetwork maps noise to NeRF weights	Works when camera poses are unknown	No explicit viewing-direction modeling	Medical DRRs (chest, knee) with unknown poses

features. pi-GAN is one of the initial models to incorporate GAN-based priors into a NeRF-like framework without explicit three-dimensional supervision.

MedNeRF model proposed by Corona-Figueroa et al. [31] is one of the first uses of NeRF in the field of medical imaging. It focuses on using a generative technique to reconstruct 3D radiographic views. The model is based on the conditional NeRF architecture and uses single-view digital radiograph (DRR) data for training. The fundamental objective of MedNeRF is to encode patient-specific information into the model to create new views of the human skull and to employ anatomical priors as a conditional radiance field.

The authors show that MedNeRF can be used effectively in medical applications even under highly constrained data regimes. However, generally it is scene-specific and struggles with generalization to other anatomical regions. Furthermore, it may not integrate advanced GAN-based style control, which limits the realism and variability of outputs. On the other hand, our proposed StyleNeRF-Med adds a modular framework that integrates StyleGAN2's latent modeling with NeRF's volumetric synthesis. This improves both anatomical variety and visual fidelity in different parts of the body.

An Unsupervised Learning of 3D Representations from Natural Images proposed by Nguyen-Phuoc et al. [29] separates posture and shape without the necessity of 3D supervision. The proposed model accomplishes this by acquiring a 3D representation within the generator, followed by a 2D projection. The fundamental concept is to depict objects in a 3D latent space, enabling the production of images from adjustable perspectives while preserving semantic consistency and realism.

The proposed model named HoloGAN is one of the fundamental models for pose-disentangled creation from a single image, although it functions within a relatively coarse 3D space, employing learned transformations instead of comprehensive volumetric rendering. Consequently, it faces challenges with intricate details and three-dimensional coherence in comparison to NeRF-based techniques. HoloGAN also relies heavily on the training dataset for posture variability.

HyperNeRF-GAN [33] is another promising model that is developed based on the NeRF and has the ability to model dynamic scenes with complex topological variations. In contrast to static NeRF-based models that presuppose a fixed scene structure, HyperNeRF integrates a deformation network that translates observations from a dynamic scene into a static space. The architecture utilizes geometry-aware GAN approaches to incorporate learned volumetric characteristics into a flexible implicit representation. This approach is particularly effective for scenes where rigid-body assumptions are inadequate, such as human motion with interacting limbs, deformable objects, or evolving environments.

A significant advancement in HyperNeRF-GAN is the implementation of continuous latent space warping, which effectively encapsulates both temporal dynamics and geometric deformation, resulting in temporally coherent and view-consistent images. The model depicts the scene as a higher-dimensional canonical space, whereby spatial and temporal elements are incorporated and visualized using volumetric rendering. Through adversarial training, the network promotes high-fidelity texture creation, while the geometry module guarantees consistent structural representation across various viewpoints and frames. This combination enables the method to surpass conventional dynamic scene reconstruction pipelines that depend on explicit meshes or surface tracking, which often struggle with occlusion and non-rigid transformations.

As summarized in Table 1, existing methods provide complementary strengths, where NeRF-based approaches ensure volumetric consistency and GAN-based models improve image realism. However, majority of these approaches are primarily designed for natural images and rely on assumptions such as dense views, clear structures, or known camera poses. Such limitations restrict their applicability to medical imaging, which involves sparse data, overlapping anatomical structures, and domain-specific variations.

3. Proposed StyleNeRF-Med model for reconstructing medical images

In this section, we present the details of our proposed approach, its motivation, system model, and detailed working of the StyleNeRF-Med which are briefly presented in this section.

3.1. Motivation and innovations of StyleNeRF-Med

Although StyleGAN2 and NeRF are both popular techniques in AI and image processing and are being used for data augmentation, image editing for virtual reality and augmented reality tasks, our proposed StyleNeRF-Med introduces an innovative integration of these models specifically for 3D medical image reconstruction using very limited 2D inputs. In contrast to existing GAN-NeRF hybrids, the proposed framework explicitly couples style-based latent modulation with volumetric radiance field learning to address sparse and ill-posed medical imaging conditions. The core novelty lies in how StyleGAN2 is used not just for texture generation but also to embed learned anatomical priors into the pipeline. These priors enhance the NeRF rendering process by enabling it to make high-quality 3D representations from sparse and noisy inputs. This particularly reduces the number of radiographs or CT scans needed for reconstruction, which can minimize the amount of radiation exposure to the patient.

Unlike prior works, which lack style-based control, the proposed StyleNeRF-Med introduces a style-modulated NeRF architecture that improves generalization across different anatomical regions. This hybrid model enables enhanced precision in controlling the output images, an aspect that has seen limited adoption in clinical settings. The results obtained show that StyleNeRF-Med consistently outperforms traditional image quality measures (FID, KID, IS) and seamlessly adapts to non-medical datasets, highlighting its potential as a universally applicable framework rather than a task-specific model.

3.2. Overview of the proposed StyleNeRF-Med

StyleNeRF-Med combines the strengths of StyleGAN2 with NeRF to achieve image reconstruction. In the proposed StyleNeRF-Med, StyleGAN2 is used to generate high-quality textures, while NeRF provides 3D volumetric representations from sparse inputs. StyleNeRF-Med addresses the challenges of medical imaging, such as limited data availability, patient movement during imaging, and overlapping anatomical structures. However, it can be generalized to other image reconstruction tasks such as creating immersive 3D environments in AR/VR for entertainment and media, improving environmental perception in robotics, reconstructing detailed 3D terrains from sparse satellite imagery, etc.

The architecture of the proposed StyleNeRF-Med is shown in Fig. 1. The StyleGAN2 generator creates new views by adapting the learned latent space. First, the latent vector is taken from a Gaussian distribution and mapped to an intermediate latent space using a mapping network. The generator then uses this latent space to generate 3D features through adaptive instance normalization (AdaIN). Once 3D features are generated, NeRF performs volumetric rendering by transforming 3D spatial coordinates and viewing directions into 2D views.

3.2.1. StyleGAN2

StyleGAN2 separates high- and low-level image attributes enabling better manipulation of features such as texture and color expressions. It manipulates the latent space and inversion to generate novel views while preserving the quality of the original images. StyleGAN2 performs a series of transformations to generate novel views, starting from a latent space and progressively refining features across layers. A vector z is sampled from a Gaussian distribution $z \sim \mathcal{N}(0, I)$ and mapped to an intermediate latent space w using a mapping network M. This process is described in Eq. (1).

$$w = M(z) \tag{1}$$

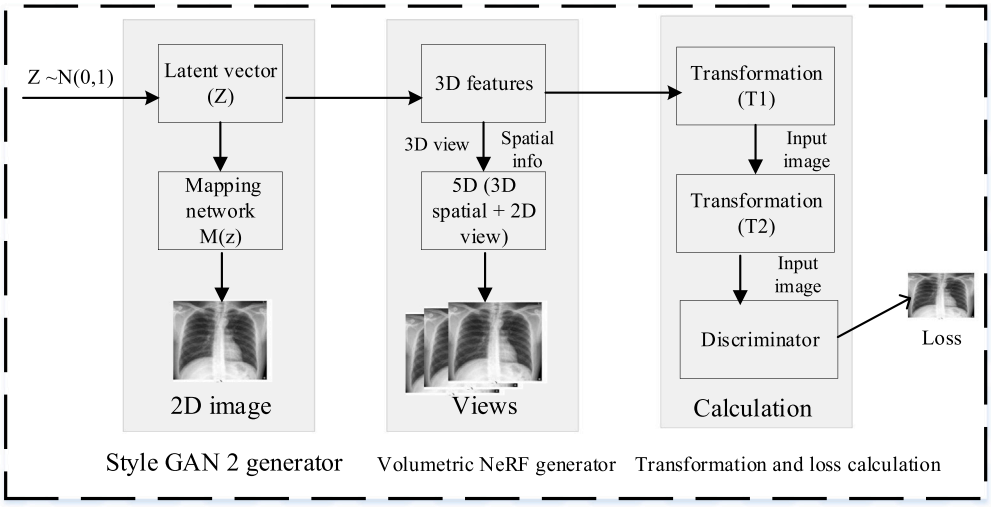


Fig. 1. Architecture of the proposed StyleNeRF-Med for medical image reconstruction.

This latent vector w is transformed to generate style parameters s_l for each layer l. AdaIN controls these style parameters as depicted in Eq. (2).

$$\text{AdaIN}(f_l, s_l) = s_{l,\text{scale}} \times \frac{f_l - \mu(f_l)}{\sigma(f_l)} + s_{l,\text{bias}} \tag{2}$$

where $\mu(f_l)$ and $\sigma(f_l)$ are the mean and standard deviation of the feature map f_l, and $s_{l,\text{scale}}$ and $s_{l,\text{bias}}$ are derived from the style parameters s_l.

The generator φ progressively refines the image, with the outputs of the layers modulated by style. At each layer l, the feature map is updated according to Eq. (3).

$$f_{l+1} = \varphi_l(\text{AdaIN}(f_l, s_l)) \tag{3}$$

The final image x is obtained after passing through all layers of the generator.

Gaussian noise ω_l is added at each layer to introduce fine-grained stochastic details, which helps in modeling subtle variations such as textures. The updated feature map is represented in Eq. (4).

$$f_{l+1} = \varphi_l(\text{AdaIN}(f_l, s_l) + \omega_l) \tag{4}$$

The discriminator D is trained to distinguish between real and fake images[1] using cross-entropy loss. The path length regularization is included to achieve smooth and consistent view generation. We present the total loss function in Eq. (5).

$$\mathcal{L}_{\text{total}} = \mathcal{L}_D + \lambda_{\text{path}} \mathcal{R}_{\text{path}} \tag{5}$$

where $\mathcal{L}_D$ is the discriminator loss, λ_{path} is a hyperparameter, and $\mathcal{R}_{\text{path}}$ represents the path length regularization term [34].

3.2.2. NeRF

The NeRF is a state-of-the-art architecture widely used for the estimating 3D volumetric representations of images. NeRF leverages a fully connected DNN for the rendering and encoding of volumetric information of complex 3D scenes. It represents 3D scenes by predicting the color and intensity of pixels in a series of 2D images by utilizing DNNs. In this way, NeRF can effectively synthesize novel views of the scene from new camera angles by querying the learned model and approximating the radiance and density at any 3D coordinate.

We train the DNN to optimize the volumetric rendering process for NeRF and create inputs in 5D coordinate systems. From these 5D coordinates, spatial coordinates are represented in 3D and viewing directions are represented in 2D. The output is the RGB color and volume intensity. To render an image, NeRF integrates these RGB colors and their densities with the rays projected from the camera into the scene.

To reconstruct a medical image from 2D to 3D, NeRF represents the image as a continuous volumetric field. The DNN F_θ employed by NeRF takes a 3D spatial coordinate $\mathbf{x} = (x, y, z)$ and a viewing direction $\mathbf{d} = (\theta, \phi)$ as input. The output consists of the RGB color $\mathbf{c} = (r, g, b)$ and the volumetric density σ, as expressed in the Eq. (6) [13].

$$F_\theta(\mathbf{x}, \mathbf{d}) = (\mathbf{c}, \sigma) \tag{6}$$

The parameterization of the rays by NeRF is defined by Eq. (7).

$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d} \tag{7}$$

where, the ray origin $\mathbf{o}$ represents the camera's position in the 3D scene, while the ray direction $\mathbf{d}$ is computed based on the camera's orientation and intrinsic parameters. These parameters are derived from known camera poses and projection matrices. t is the distance along the ray.

The contributions of points sampled along the ray are collected and their integrals are performed with the density σ and radiance $\mathbf{c}$ to obtain the color of the pixel $\hat{C}(\mathbf{r})$. This is calculated as in Eq. (8) [13].

$$\hat{C}(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\sigma(\mathbf{r}(t))\mathbf{c}(\mathbf{r}(t), \mathbf{d})\, dt \tag{8}$$

The integral in Eq. (8) is computed numerically using discrete sampling along the ray. Specifically, the interval $[t_n, t_f]$ is divided into evenly spaced samples N, and the integral is approximated as a weighted sum of the radiance and density values at these points.

The accumulated transmittance $T(t)$ denotes the probability that light travels along the ray $\mathbf{r}(t) = \mathbf{o} + t\,\mathbf{d}$ (see Eq. (7)) without being absorbed between the near/far bounds t_n and t_f. Let $\sigma(\mathbf{x})$ be the volumetric density at 3D point $\mathbf{x}$. We define the optical depth and the resulting transmittance as in Eq. (9).

$$\begin{aligned} \tau(t) &= \int_{t_n}^{t} \sigma\big(\mathbf{r}(s)\big)\, ds, \\ T(t) &= \exp\big(-\tau(t)\big). \end{aligned} \tag{9}$$

[1] A "fake" image refers to an image synthesized by the generator, as opposed to a "real" image, which comes from the actual dataset.

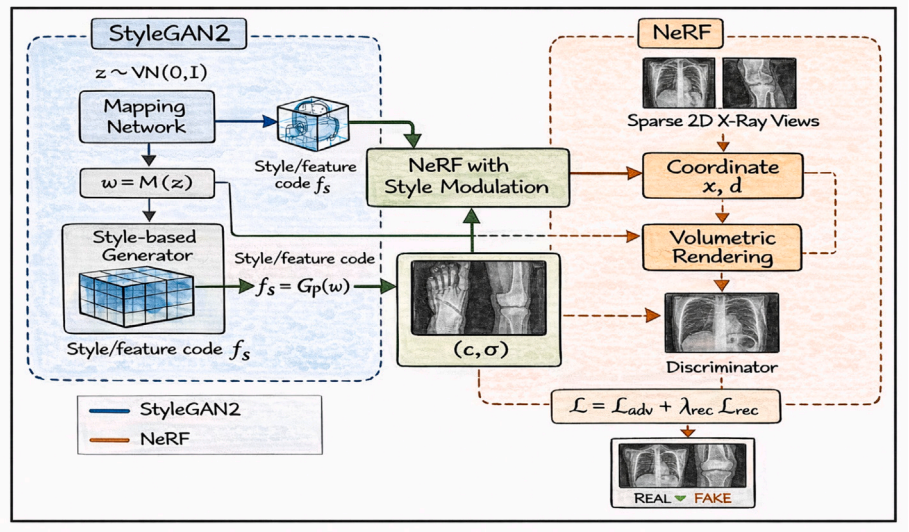


Fig. 2. Integration of StyleGAN2 and NeRF in StyleNeRF-Med.

To approximate the volumetric integral, the ray is discretized into N samples. The discrete approximation for the color of the pixel is represented in Eq. (10) [13].

$$\hat{C}(\mathbf{r}) \approx \sum_{i=1}^{N} T_i(1 - \exp(-\sigma_i \Delta t_i))\mathbf{c} \tag{10}$$

where T_i is the accumulated transmittance up to the i-th sample, Δt_i is the distance between consecutive samples, σ_i is the density at the i-th point and $\mathbf{c}_i$ is the color at the same point. The network approximates this by predicting the radiance $\mathbf{c}_i$ and density σ_i, which are used to calculate the opacity α_i and transmittance T_i, determining the weighted contribution of each sample to the final color.

The network is trained by minimizing the loss function between the predicted color $\hat{C}(\mathbf{r})$ and the ground truth color $C(\mathbf{r})$ in all rays for the training set. The loss function is given in Eq. (11).

$$\mathcal{L} = \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}\in\mathcal{R}} \left\| \hat{C}(\mathbf{r}) - C(\mathbf{r}) \right\|^2 \tag{11}$$

where $\mathcal{R}$ denotes the set of all rays in the dataset.

3.2.3. Integration of StyleGAN2 and NeRF

StyleNeRF-Med integrates StyleGAN2 and NeRF in a unified and end-to-end trainable pipeline. The mapping network of StyleGAN2 maps the latent vector z to an intermediate latent space w which is then modulated by the generator to produce structured feature representations. These features are incorporated into the NeRF alongside 3D spatial coordinates (x, d) and viewing directions, enabling radiance and density prediction guided by style-aware latent information. The NeRF model thus performs style-conditioned volumetric rendering to generate novel views. The entire framework is trained jointly in an end-to-end manner, enabling effective feature exchange where StyleGAN2 provides global structural priors and NeRF ensures spatial consistency through volumetric modeling. The integration of StyleGAN2 and NeRF in StyleNeRF-Med is shown in Fig. 2.

3.3. Working of the proposed StyleNeRF-Med model

In this subsection, we briefly describe the workings of the proposed StyleNeRF-Med for medical image reconstruction. We present the pseudocode for the algorithm for StyleNeRF-Med as an Algorithm 1.

The Algorithm 1 presents how StyleNeRF-Med integrates StyleGAN2 and NeRF to generate new views for medical images. At the start, the algorithm chooses a latent vector (z) from a Gaussian distribution and transforms it into an intermediate latent space (w) using a mapping network (M). Afterwards, it creates the Style parameters (s_l) and adjusts the layers of the generator. Then the NeRF performs volumetric rendering by estimating radiance $\mathbf{c}$, density σ and pixel colors.

Training losses, both for the generator and the discriminator, are calculated by comparing the real images from the dataset and the generated images. A backpropagation mechanism is used that allows StyleGAN2 and NeRF to be updated iteratively. The proposed StyleNeRF-Med leverages the capabilities of StyleGAN2 for rendering high quality image instances and the power of NeRF for modeling 3D scenes to synthesize new views for medical images.

The Algorithm 2 provides details of the backpropagation and update procedures executed during the training phase to improve understanding of the optimization process. It delineates the process by which the generator and NeRF networks are refined by minimizing generator loss, while the discriminator is improved by optimizing adversarial

Algorithm 1 Pseudocode for the proposed StyleNeRF-Med.

Input: Dataset D with image-ground truth pose pairs (I, P), StyleGAN2 generator φ, NeRF network F_θ
Output: Generation of novel views $\hat{I}$
Initialize φ (StyleGAN2 Generator and Discriminator)
Initialize F_θ (NeRF for volumetric rendering)
Load dataset $D = \{(I_k, P_k)\}_{k=1}^{N}$, where I_k are images and P_k are camera poses
for each training step **do**
    Sample a latent vector $z \sim \mathcal{N}(0, I)$
    Sample a batch of image-pose pairs (I, P) from D
    // Generate synthetic images
    $w \leftarrow M(z)$
    $\{s_l\} \leftarrow \varphi(w)$
    // Perform volumetric rendering with NeRF
    $(\mathbf{c}, \sigma) \leftarrow F_\theta(\mathbf{x}, \mathbf{d})$
    $\hat{C}(\mathbf{r}) \leftarrow \sum_{i=1}^{N} T_i(1 - \exp(-\sigma_i \Delta t_i))\mathbf{c}_i$
    // Calculate losses and optimize
    $\mathcal{L}_G \leftarrow \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}\in\mathcal{R}} \|\hat{C}(\mathbf{r}) - C(\mathbf{r})\|^2$
    $\mathcal{L}_D \leftarrow \text{CrossEntropy}(D(\hat{I}), \text{Real/Fake})$
    Backpropagate and update networks φ, F_θ, and D
end for

Algorithm 2 Optimization and steps for StyleNeRF-Med.

Input: Losses $\mathcal{L}_G$, $\mathcal{L}_D$; networks φ, F_θ, D; latent vector z
Output: Updated parameters for φ, F_θ, D, and optionally z

// Backpropagate generator loss through StyleGAN2 and NeRF
Compute gradients of $\mathcal{L}_G$ w.r.t. parameters of φ and F_θ
Update parameters of φ and F_θ using optimizer (e.g., Adam)
// Backpropagate discriminator loss
Compute gradients of $\mathcal{L}_D$ w.r.t. parameters of D
Update parameters of D using optimizer
// Latent space update (if latent optimization is used)
if latent optimization is enabled **then**
    Compute gradients of $\mathcal{L}_G$ w.r.t. latent vector z
    Update z using optimizer
end if
// Discriminator role
Discriminator D receives real images and generated images $\hat{I}$, outputs probability of real/fake, and is updated via $\mathcal{L}_D$

loss. The pseudocode highlights the optimization of the latent space, whereby the latent vector is modified. During each training iteration, the discriminator is presented with real and generated images, evaluates the probability of authenticity, and is subsequently updated. This detailed account guarantees openness in the training process and clarifies the function of each network component within the overarching architecture.

4. Implementation details

We implement the StyleNeRF-Med model for the reconstruction of medical images. We use a PC equipped with powerful GPUs for the implementation of the model. All the training is done on an Nvidia RTX-4070 GPU with 12GB memory. We utilize the publicly available source code for different models such as GRAF[2] [16], PixelNeRF[3] [18], and HyperNeRFGAN[4] [33]) to implement our approach. We re-train all these models with our processed datasets, and parameters, as described in the following subsections.

4.1. Dataset preparation

For training purposes, we synthesize the medical image data and generate Digitally Reconstructed Radiographs (DRRs) using the similar approach to that used by Corona-Figueroa et al. [31]. The use of synthetically generated DRRs enables controlled conditions for evaluating reconstruction performance under sparse-view settings. DRRs preserve important structural and anatomical characteristics of medical imaging while allowing systematic variation in projection geometry. This enables consistent and reproducible evaluation, which is difficult to achieve with limited real clinical datasets. The DRR-based evaluation approach is common and established practice in medical image reconstruction research, where models are first validated under controlled synthetic conditions before being extended to real clinical data [31]. We generate DRRs from 20 chest CT scans from sources [35,36] and denote this dataset by D_{chest}. The other DRRs are generated by utilizing knee CT scans of five patients from sources [37,38] and are denoted as D_{knee}. These scans represent a diverse group of patients who exhibit a variety of contrast types, illustrating both pathological and normal anatomy. The radiation source and imaging panel are assumed to rotate around the vertical axis. This assumption leads us to generate 128×128 DRRs, with each DRR taken at 5°. To complete the 360° view, a total of 72 DRRs have been generated for both chest and knee datasets. 72 images of each D_{chest} and D_{knee} thus represent one-fifth of the 360° rotation of complete possible views. We present the details of both datasets in Table 2. The generated DRR images are split into 80% training data and 20% testing data for model development and unbiased evaluation.

Furthermore, for testing the scalability aspect, we increased the dataset size and introduced resolution-based diversity, we doubled the dataset size by downsampling from 128×128 to 64×64 image size. This effectively doubled the number of training samples and allowed the model to learn from multiple input resolutions.

Table 2
Details of datasets used.

Dataset	Size	Description
D_{chest1}	128×128	1440 DRRs from 20 patients, each DRR is at 5°
D_{knee1}	128×128	360 DRRs from 5 patients, each DRR is at 5°
D_{chest2}	64×64	1440 DRRs from 20 patients, each DRR is at 5°
D_{knee2}	64×64	360 DRRs from 5 patients, each DRR is at 5°

4.2. Methodology

We train our proposed StyleNeRF-Med model to reconstruct medical images leveraging StyleGAN2 and NeRF up to 100,000 iterations. StyleGAN2 in StyleNeRF-Med generates 2D images while retaining the intricate details of the input data, whereas NeRF is used to generate 3D renderings for volumetric reconstructions of novel views of medical images.

The entire framework is trained jointly in an end-to-end manner, where both StyleGAN2 and NeRF components are optimized simultaneously using combined adversarial and reconstruction losses. We employ hypernetwork to set the hyperparameters for the training process. Setting hyperparameters using hypernetwork is a well-known approach and is widely used [39–41]. The hypernetwork dynamically predicts and adaptively updates parameters according to the current training state. In addition to adaptive hyperparameter setting, we also set the static parameters to form the baseline for architectural configurations. The static parameters are shown in Table 3.

To tackle the challenge of model overfitting, we employ standard regularization and augmentation techniques within the training pipeline [42]. Particularly, data augmentation is applied using an adaptive augmentation pipeline that includes geometric transformations, color intensity variations, and horizontal flipping. Furthermore, GAN-specific

Table 3
Hyperparameter configuration of the proposed StyleNeRF-Med framework.

Hyperparameter	Value
StyleGAN2	
Generator mapping layers	2
Generator feature maps	1
Generator input resolution	4
Noise injection	Enabled
Discriminator feature maps	1
R1 regularization weight	10.0
NeRF	
Generator network width	128
Generator number of layers	4
Samples per ray	32
Perturbation factor	1.0
Discriminator feature maps	1
R1 regularization weight	0.01
Training	
Batch size	8
Training iterations	100,000

[2] https://github.com/autonomousvision/graf.
[3] https://github.com/sxyu/pixel-nerf.
[4] https://github.com/gmum/HyperNeRFGAN.

regularization techniques are incorporated, including the R1 gradient penalty for the discriminator and path-length regularization for the generator.

4.3. Evaluation metrics

The proposed StyleNeRF-Med is designed for generative novel view synthesis; therefore to evaluate its performance, we adopt well known evaluation metrics widely used for the assessment of generative models. These metrics include Fréchet Inception Distance (FID), Kernel Inception Distance (KID), and Inception Score (IS), which are used to evaluate the visual quality and realism of the generated images [27,30]. We limit our evaluation to the generative capabilities; thus the additional metrics for pixel-wise reconstruction such as PSNR and SSIM are beyond the scope of this study.

FID measures the difference between the feature distributions of real and generated images by comparing their mean (μ) and covariance (υ) using Eq. (12).

$$FID = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}(\upsilon_r + \upsilon_g - 2\sqrt{\upsilon_r \upsilon_g}) \tag{12}$$

where (μ_r, υ_r) and (μ_g, υ_g) represent the characteristic statistics of the real and generated images, respectively.

KID performs the calculation analogously to FID, but uses a kernel-based approach and computes the squared Maximum Mean Discrepancy (MMD) as in Eq. (13).

$$KID = \mathrm{average}\left[k(x, x') + k(y, y') - 2k(x, y)\right] \tag{13}$$

where $k(x, y)$ is a polynomial kernel that compares features of real images (x) and generated images (y). The lower values of FID and KID reflect better performance. IS quantifies both realism and diversity of generated images by measuring how confident a pre-trained classifier is about each image [30].

$$IS = \exp\left(\mathbb{E}_{x\sim p_g}\left[D_{\mathrm{KL}}\left(p(y \mid x) \,\|\, p(y)\right)\right]\right) \tag{14}$$

where $p(y \mid x)$ is the class-probability vector predicted by an inception network for a generated image x, and $p(y)$ is the marginal class distribution across all generated samples.

For fair comparison, all baseline models were trained and evaluated on the same processed datasets under the same train/test split and input resolutions used for the StyleNeRF-Med.

5. Results and evaluation

After describing the evaluation strategy, we present a visualization of the output of the proposed StyleNeRF-Med for the chest and knee datasets in Figs. 3 and 4, respectively. For a complete 360-degree vertical rotation, the projection parameters (u, v) are chosen such that the positions on the surface of a sphere are uniformly sampled. Specifically, a small horizontal elevation of 70°-85° and $u_{min} = 0$ and $u_{max} = 1$ are specified. The results show the ability of our proposed model to generate 3D-capable DRRs by using a single-view X-ray image as input. Our model is able to accurately represent the attenuation response and 3D anatomical identity of a variety of medical cases. This shows that the model can accurately interpret the attenuation of X-rays by different tissues and effectively distinguish different anatomical structures in three dimensions.

5.1. Quantitative evaluation

Once we show that the proposed StyleNeRF-Med reconstructs medical images, we provide benchmarking with existing models and quantitatively evaluate the quality of the generated images. The results are presented in Figs. 5–10.

The results in Figs. 5–7 show the values obtained for FID, KID, and IS at the resolutions 64×64 and 128×128 for the chest dataset. The values show that the proposed StyleNeRF-Med model outperforms all other models in all performance parameters. For the 64×64 dataset, it achieved lower FID/KID values (103.13 / 0.07) than the other models, while achieving a higher IS score (1.94). The same trend is preserved when the image size is changed to 128×128.

Similarly, Figs. 8–10 present the comparison that focuses on FID, KID, and IS for the Knee dataset. For the knee dataset, at 64×64, the proposed model achieves FID/KID (152.23/0.11) and IS (2.49), which are better than the other models. When the image size of the dataset is 128×128, the proposed model again clearly outperforms all existing models in all three parameters.

5.2. Ablation study and computational complexity analysis

To evaluate the effectiveness of the proposed StyleNeRF-Med, we also provide a systematic evaluation through an ablation study. We consider the component-level analysis that compares NeRF components, followed by a full model analysis. We also analyze the scalability aspect when the dataset is doubled. Component-level analysis shows that for the chest dataset at 64×64 resolution, the NeRF in StyleNeRF-Med requires 16.71 ms inference time and 5.89 MB of memory compared to the standard NeRF's 1.87 ms and 4.34 MB. This pattern is consistent across both datasets and resolutions. For the full model analysis, we observe consistent performance characteristics across datasets. In the chest dataset, StyleNeRF-Med achieves an inference time of 19.16 ms at 64×64 resolution compared to NeRF's 11.19 ms, requiring 1829.20 MB memory versus NeRF's 898.52 MB. At 128×128 resolution, StyleNeRF-Med

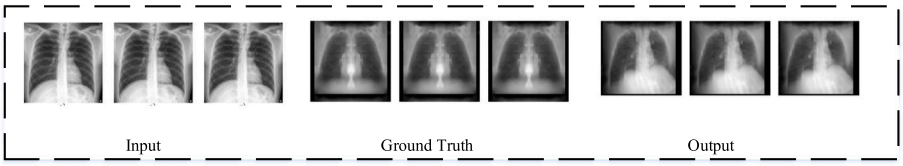


Fig. 3. Generated images from chest X-rays.

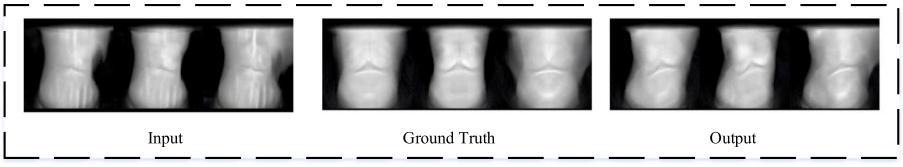


Fig. 4. Generated images from knee X-rays.

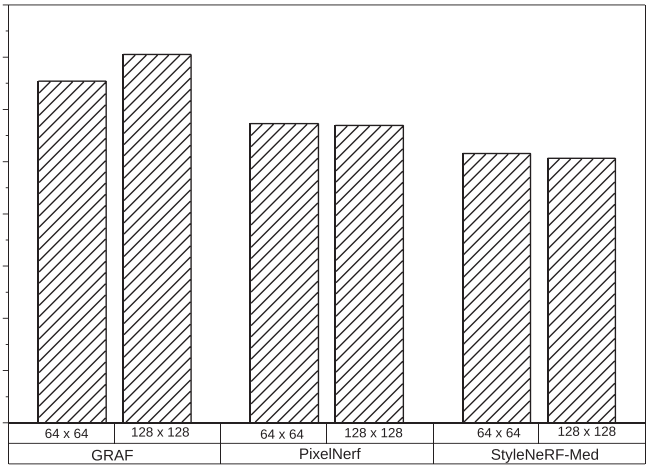


Fig. 5. FID values for chest dataset.

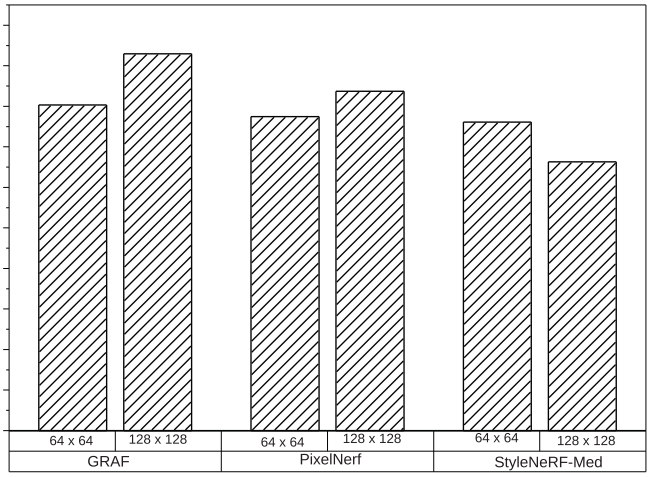


Fig. 8. FID values for Knee dataset.

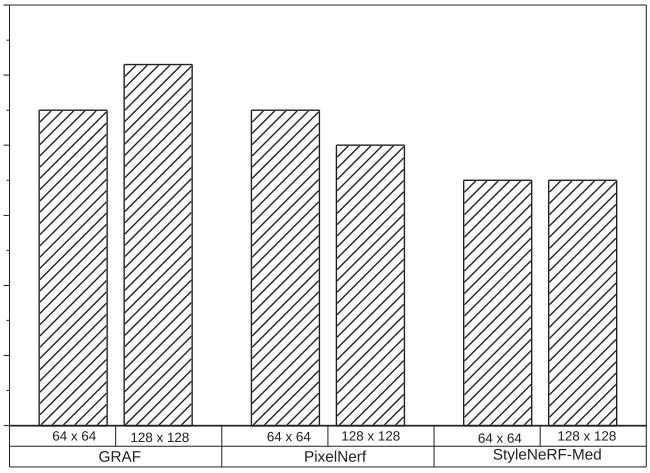


Fig. 6. KID values for Chest dataset.

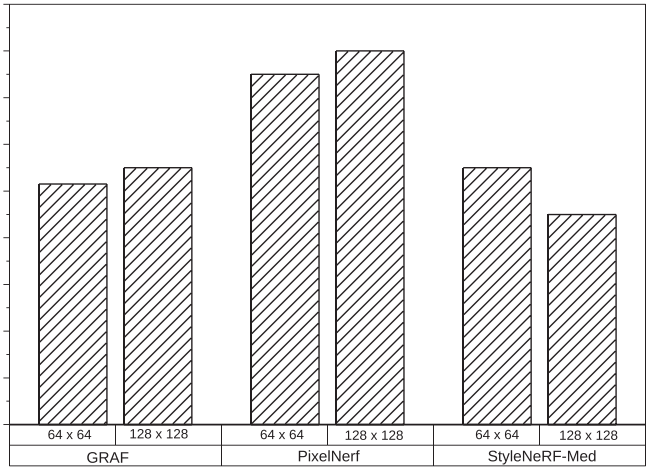


Fig. 9. KID values for Knee dataset.

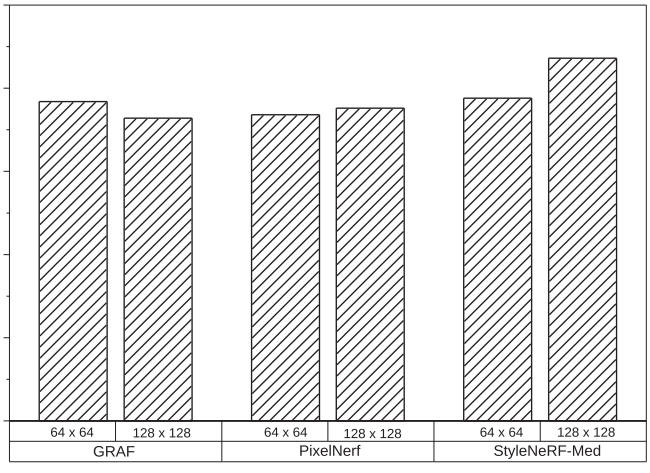


Fig. 7. IS values for Chest dataset.

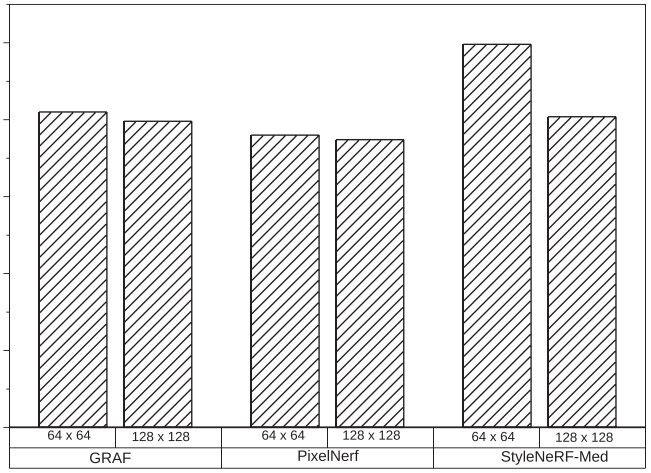


Fig. 10. IS values for knee dataset.

shows an inference time of 20.96 ms with 1832.34 MB memory compared to NeRF's 12.61 ms and 901.66 MB, demonstrating the expected computational overhead. In the knee dataset, both models show similar performance patterns at 64×64 resolution (19.20 ms vs 10.79 ms inference time), with StyleNeRF-Med maintaining consistent patterns of memory usage. We summarize the analysis at the component level and the full model in Tables 4 and 5, respectively.

Table 4
Component-Level analysis (NeRF components only).

Dataset	Resolution	Model	Inference Time (ms)	Memory Used (MB)
Chest	64 × 64	StyleNeRF-Med	16.71	5.89
		NeRF	1.87	4.34
	128 × 128	StyleNeRF-Med	9.83	5.89
		NeRF	1.75	4.34
Knee	64 × 64	StyleNeRF-Med	9.86	5.89
		NeRF	1.84	4.34
	128 × 128	StyleNeRF-Med	8.73	5.89
		NeRF	1.67	4.34

Table 5
Full model Analysis.

Dataset	Resolution	Model	Inference Time (ms)	Memory Used (MB)
Chest	64 × 64	StyleNeRF-Med	19.16	1829.20
		NeRF	11.19	898.52
	128 × 128	StyleNeRF-Med	20.96	1832.34
		NeRF	12.61	901.66
Knee	64 × 64	StyleNeRF-Med	19.20	1829.20
		NeRF	10.79	898.52
	128 × 128	StyleNeRF-Med	20.78	1832.34
		NeRF	12.43	901.66

Lastly, we also analyze the effect of scalability by doubling the size of the dataset from 64×64 to 128×128 image sizes. The analysis reveals that scaling from 64×64 to 128×128 resolution affects both models similarly in terms of inference time but shows realistic memory scaling. For the chest dataset, StyleNeRF-Med shows a 1.1 × increase in inference time (19.16 ms 20.96 ms) with a proportional memory increase (1829.20 MB →1832.34 MB), while NeRF shows a 1.1 × increase in inference time (11.19 ms →12.61 ms) with a proportional memory increase (898.52 MB →901.66 MB). For the knee dataset, both models show similar scaling patterns.

The integration of StyleGAN2 and NeRF naturally introduces additional computational overhead compared to standard NeRF-based baselines, due to the joint optimization of style-based generation and volumetric rendering. The increased computational overhead of StyleNeRF-Med is fundamentally justified by its unique architectural capabilities that standard NeRF cannot provide. Unlike reconstruction-based NeRF approaches that can only recover existing 3D scenes from input views, StyleNeRF-Med enables novel 3D content generation through learned latent space manipulation. This paradigm shift from reconstruction to generation provides unprecedented value for medical diagnostics such as generating diverse anatomical variations, and synthesizing missing views that standard NeRF based models are unable to accomplish.

5.3. Generalization of StyleNeRF-Med to non-medical use cases

We also demonstrate the capabilities of the proposed StyleNeRF-Med for non-medical use cases. We run StyleNeRF-Med in non-medical domains by evaluating its performance with the publicly available ShapeNet cars dataset[5]. ShapeNet contains a collection of various high-quality 3D models. We used the car dataset due to its large structural and visual diversity. The results are shown in Fig. 11, which shows the ability of the proposed model to generate new high-quality views.

[5] https://github.com/gmum/HyperNeRFGAN.

The StyleNeRF-Med architecture shows robust performance in this non-medical environment. This observation is consistent with recent multi-view representation learning research in non-medical imaging domains, where self-supervised learning has been used to improve fine-grained recognition from multi-view infrared images [43]. The model effectively creates visually accurate 360 degree car views using StyleGAN2 for latent space manipulation and NeRF for volumetric rendering. The model's ability to separate and retain key features in latent space ensures that the results maintain a high degree of realism, even when applied to a dataset whose features differ significantly from those of the medical domain. After demonstrating the capabilities of proposed StyleNeRF-Med on ShapeNet car data, we briefly present some other non-medical potential use cases for StyleNeRF-Med.

5.3.1. Cultural heritage and archaeology

StyleGAN2–NeRF can recreate three-dimensional models of artifacts, sculptures, or historical places using a limited number of two-dimensional photos. This is valuable when physical access is limited, facilitating digital preservation, virtual museums, and instructional uses.

5.3.2. Entertainment and creative industries

NeRF-based models are already being utilized in visual effects, gaming, and AR/VR, where high-fidelity 3D materials are required from a limited number of images. StyleGAN2 enhances controllability, facilitating artistic modifications and domain adaptation (e.g., transforming sketches into 3D characters).

5.3.3. Robotics and autonomous navigation

Synthetic 3D sceneries derived from sparse 2D images can facilitate robot simulation, environmental comprehension, and secure path planning in scenarios where comprehensive real-time 3D mapping is impractical.

5.4. Discussion

In this work, we propose StyleNeRF-Med, which innovatively integrates StyleGAN2 and neural radiance fields (NeRF) to generate novel 360-degree views of medical images with fewer input data. The proposed work reduces the reliance on multiple, often invasive or radiation-intensive medical images to create detailed anatomical reconstructions to facilitate disease diagnosis. By synthesizing high-quality medical images from a smaller amount of input data, StyleNeRF-Med has the potential to significantly reduce patient exposure to harmful radiation (e.g., X-rays and CT scans) and streamline diagnostic workflows—providing significant clinical impact.

The proposed StyleNeRF-Med aims to capture fine-grained anatomical structures and ensure that clinically relevant features are preserved. We implement the proposed model using digitally reconstructed radiographs (DRRs) obtained from chest and knee—two regions where precise anatomical details are crucial. The obtained results and their quantitative evaluation, measured by Fréchet Inception Distance (FID) Kernel Inception Distance (KID), and Inception Score (IS) show significant improvements over existing models, highlighting the state-of-the-art performance of StyleNeRF-Med. We also performed an ablation study on the training iterations to further illustrate that the model benefits from extended training, with FID values improving significantly, confirming the model's learning ability and convergence behavior.

The majority of existing 3D reconstruction models are either based on GANs with limited view consistency or use NeRF without style-based control, which often leads to poor generalization. The proposed StyleNeRF-Med combines StyleGAN2 style control with NeRF view synthesis capabilities to produce anatomically coherent reconstructions with better generalizability. To validate the generalizability, we also implement our model on the non-medical ShapeNet cars dataset. The model successfully generates consistent and realistic views, highlighting its adaptability across different domains and data modalities. This

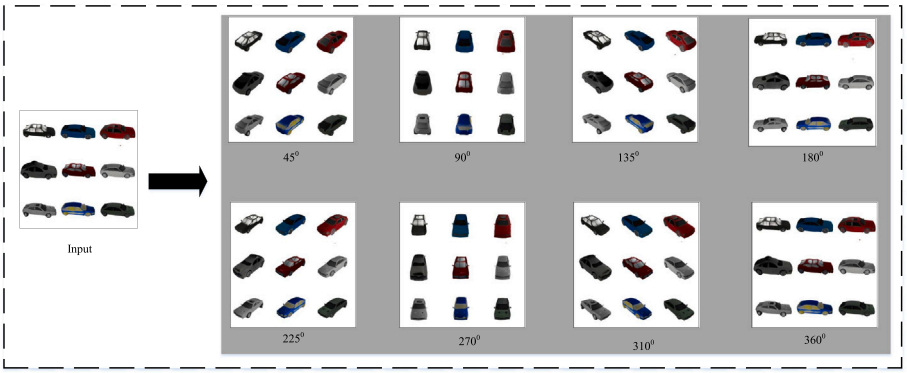


Fig. 11. Generation of 360 degree views on non-medical data.

cross-domain performance demonstrates the potential of StyleNeRF-Med as a generic framework for 3D image synthesis.

The proposed StyleNeRF-Med is designed to operate effectively under sparse-view conditions, where only a limited number of input images are available. In this work, we evaluate the model using limited-input settings and further assess its behavior at two image resolutions, 64×64 and 128×128, to examine robustness under different input scales. The results show consistent performance across these settings, indicating the stability of the proposed framework under constrained data conditions. A more detailed sensitivity analysis with varying numbers of input views will be explored in future work.

5.5. Limitations and future work

In this study, we adopt synthetically generated DRR data as a controlled and widely used preliminary evaluation setting for medical image reconstruction. DRR-based experimentation enables consistent analysis of sparse-view reconstruction behavior while preserving important anatomical structures and imaging geometry. Although this setting is commonly used as an initial step in the literature, some differences may still exist between synthetic DRRs and real clinical images, for example in noise characteristics and imaging artifacts. Therefore, we consider DRR-based evaluation as a complementary stage for model development, repeatability, and algorithm stress-testing rather than a replacement for real clinical validation. In future work, we plan to extend the proposed framework to real clinical datasets, evaluate its performance under varying numbers of input views, and obtain expert assessments of anatomical fidelity from clinical professionals such as radiologists.

6. Conclusion

Reconstructing internal anatomy from a few medical images (X-rays or scans) is crucial to an accurate prognosis of the disease. Existing image reconstruction techniques, such as CT and magnetic resonance imaging (MRI), often require multiple medical images to produce an accurate image. However, acquiring multiple images may lead to an extended exposure of patients to ionizing radiation for longer periods of time, which in turn is hazardous to the patient's health. In this work, we propose StyleNeRF-Med, a novel approach to reconstruct medical images in high quality by integrating Generative Adversarial Networks (GANs) with NeRF. The proposed StyleNeRF-Med leverages the strong capabilities of StyleGAN2 to generate detailed views by encoding various image features. The NeRF in StyleNeRF-Med processes 3D spatial coordinates and camera positions to accurately compute pixel colors and intensities in 3D space. We conducted the experiments by implementing StyleNeRF-Med with Digitally Reconstructed Radiographs (DRRs). The results show that the proposed StyleNeRF-Med effectively generates new views for medical images with fewer input images. We also compare the quality of StyleNeRF-Med with the existing models. The evaluation shows that StyleNeRF-Med outperforms the existing models by generating high-quality images.

CRediT authorship contribution statement

Syed Muhammad Abrar Akber: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Sadia Nishat Kazmi:** Writing – review & editing, Visualization, Methodology, Formal analysis. **Ali Muqtadir:** Visualization, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have provided references for the dataset used.

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